IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2020

**ANALYSIS AND DESIGN OF CIRCUITS**

Wednesday, 29 April 10:00 am

Time allowed: 2:00 hours

**There are THREE questions on this paper.**

**Answer ALL questions. Question One carries 40% of the marks. Question 2 and 3 carry equal marks (30% each)**

**Any special instructions for invigilators and information for candidates are on page 1.**

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Second Marker(s) : P.D. Mitcheson, A.S. Holmes

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## Information for Candidates

### Numerical answers

- Numerical answers should be given as fully evaluated decimal values and not as unevaluated arithmetic expressions.

### Notation

The following notation is used in this paper:

- Component and source values can be assumed to be in Ohms, Farads, Henrys, Volts or Amps where unit symbols are omitted. Where an imaginary number is specified, it represents the complex impedance or phasor value.

- Times are in seconds unless otherwise stated.


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**The Questions**

1. a) Using nodal analysis, calculate the voltages at nodes *X* and *Y* of Figure 1.1.

[6]

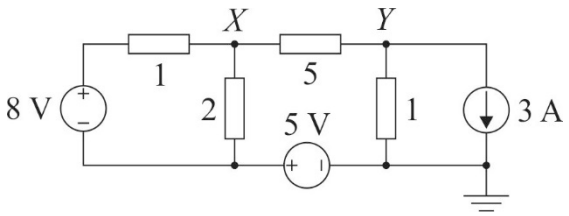



*Figure 1.1*

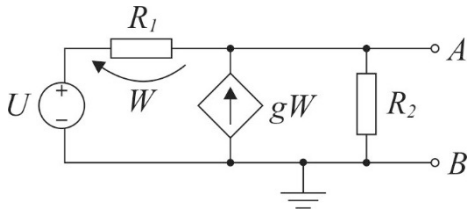



*Figure 1.2*

b) Draw the Thévenin equivalent of the network shown in Figure 1.2 and derive expressions for the values of its components. Note that the network contains a voltage-controlled current source.

[6]

c) For the circuit in Figure 1.3, write down an expression relating *Z* to *X* and *Y*. You should assume the opamp is ideal.

[5]

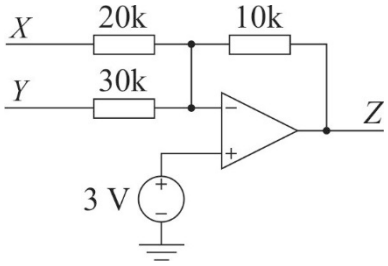



*Figure 1.3*

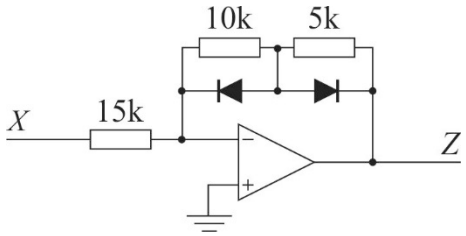



*Figure 1.4*

d) In the circuit of Figure 1.4, the diodes have a forward voltage of 0.7 V when conducting but are otherwise ideal. Draw a dimensioned sketch showing the variation of *Z* with *X* for input voltages in the range $-3 \le X \le 3$.

[6]

**Question 1 continues on the next page…**

**Question 1 continued**

e) For the circuit in Figure 1.5, choose the value of $R_B$ to give a bias voltage of 2.5 V at the collector of the transistor. [5]

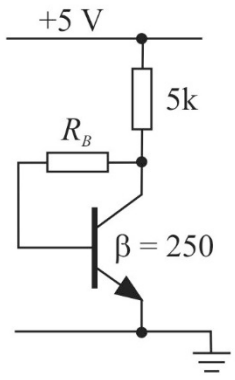



*Figure 1.5*

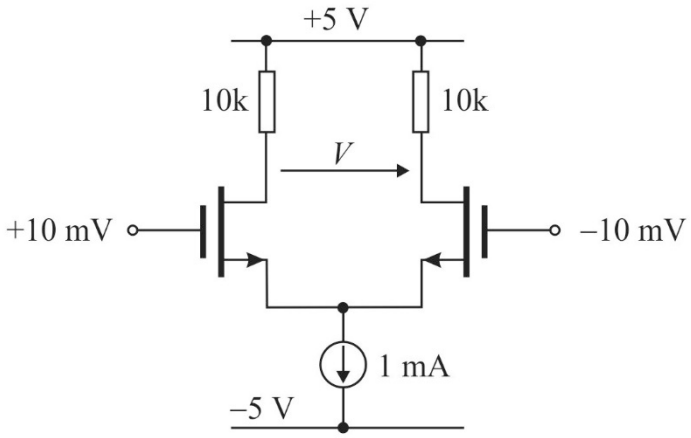



*Figure 1.6*

f) The MOSFETs in Figure 1.6 are matched, with $K = 0.5\ \text{mA/V}^2$. Calculate the differential output voltage, *V*, assuming the MOSFETs have infinite output resistance. [6]

g) For the partial circuit in Figure 1.7, assuming the drain of the MOSFET is held at a potential such that the device is in saturation, show that the drain current is 0.1 mA. Also calculate the small-signal output resistance under these conditions. [6]

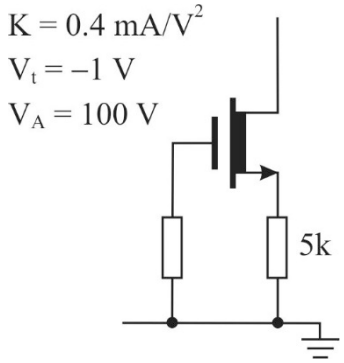



*Figure 1.7*

2. Figure 2.1 shows a simple power amplifier which is to be used to drive a loudspeaker. The amplifier comprises an inverting op-amp circuit with a transistor output stage to boost the output power capability.

   a) If the opamp stage is required to have an input impedance of 1 kΩ and a voltage gain of −20 in the mid-band, choose values for $R_1$ and $R_2$.

   [6]

   b) If the opamp stage is required to have a band-pass characteristic with lower and upper corner frequencies of 20 Hz and 20 kHz, choose values for $C_1$ and $C_2$. Draw a dimensioned sketch showing the magnitude frequency response of the opamp stage in this case.

   [10]

   c) The amplifier is tested and the audio quality is found to be poor. Explain why this is the case. Illustrate your answer by sketching the output waveform you would expect to see when a 1 kHz sine wave of amplitude 200 mV peak-to-peak is applied at the input.

   [8]

   d) Sketch an alternative output stage configuration that would give improved audio quality. Can you also see a way to improve the audio quality that does not involve adding any components?

   [6]

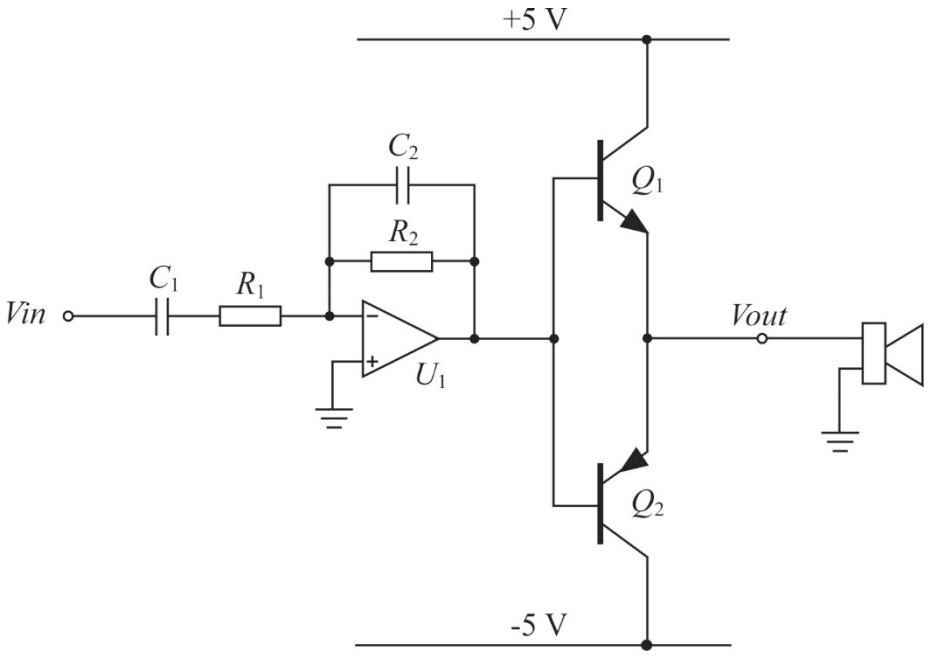



*Figure 2.1*

3. Figure 3.1 shows a common emitter amplifier with an active load. All three transistors have $\beta = 100$ and $V_A = 150$ V.

a) Ignoring base currents, calculate the value of $R_2$ that will cause the transconductance of $Q_1$ to be 40 mS.

[2]

b) Calculate the value of $R_1$ that will bias the collector voltage of $Q_1$ to achieve equal head and leg room.

[2]

c) Draw a small-signal equivalent model of the amplifier and calculate the values of any small-signal parameters. (Hint: $Q_2$ and $Q_3$ only contribute one resistor to the SSEM.)

[6]

d) Calculate the small-signal input resistance $R_i$, output resistance $R_o$ and voltage gain $A_v$ of the amplifier in the mid-band.

[6]

e) i) Calculate the corner frequency of the high-pass filter associated with $C_{in}$ when the amplifier is driven by a source with resistance 50 Ω.

[3]

ii) Calculate the corner frequency of the high-pass filter associated with $C_{out}$ when the amplifier is driving a 50 Ω load.

[3]

f) A 10 pF capacitor, $C_1$, is placed in parallel with $R_1$. Calculate the apparent input capacitance $C_1$' between the base and ground due to this capacitance. You should use the $A_v$ value you calculated in part d) in this calculation.

[4]

g) By considering the network formed by $R_i$, $C_1$' and the source resistance, calculate the corner frequency of the low-pass filter associated with $C_1$ when the amplifier is driven by a source with resistance 50 Ω. You should assume that $C_{in}$ has zero impedance for this calculation.

[4]

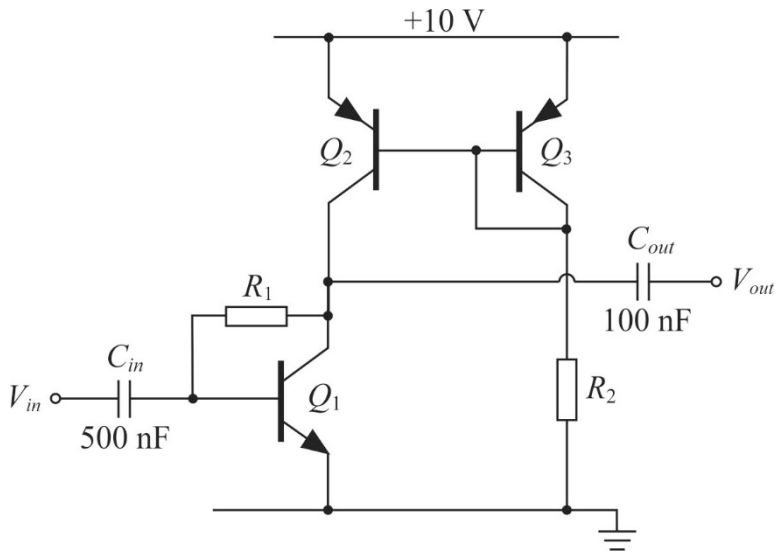



*Figure 3.1*